



CONTROLLED DRAFT

# QA, Device Matrix, Release & Signing Plan

Validation strategy from transport spike through staged Google Play rollout

|                  |                            |
|------------------|----------------------------|
| Product          | Nod                        |
| Document version | 1.0                        |
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| Owner / operator | [NOD OWNER / LEGAL ENTITY] |
| Status           | Controlled Draft           |

**Release note:** This pack is a production-ready baseline. Bracketed operator details must be completed before public release. Legal documents should be reviewed by Nigerian counsel before publication or execution.

## 1. Test layers

| Layer           | Coverage                                                                                              |
|-----------------|-------------------------------------------------------------------------------------------------------|
| Unit            | State reducers, protocol encoding, validation, shake algorithm, phone normalization, hashing helpers. |
| Integration     | Room repositories, DataStore, transport adapter contract tests, file temp/finalization.               |
| UI              | Compose navigation, permission denial/retry, manifest/consent, errors, accessibility.                 |
| Physical device | Wireless discovery, pairing, background, OEM battery behavior, large transfer.                        |
| Security        | Malformed input, spoofed alias, replay/duplicate messages, hash mismatch, hostile filenames.          |
| Performance     | Startup, discovery latency distribution, transfer throughput, memory, battery, large files.           |

## 2. Minimum physical matrix

- Samsung current + older supported model.
- Tecno model common in Nigerian market.
- Infinix model common in Nigerian market.
- Xiaomi/Redmi.
- Google Pixel reference device.
- At least one API 26/27 or earliest practically obtainable supported device.
- At least one device on current target-generation Android.

## 3. Pair matrix

- Samsung <-> Samsung
- Samsung <-> Tecno
- Tecno <-> Infinix
- Xiaomi <-> Samsung
- Pixel <-> Tecno
- Different Android major versions
- GMS capability present/absent detection path

## 4. Stress scenarios

- 100 KB, 100 MB, 1 GB and multi-GB payloads.
- Multiple mixed files.
- Cancel by sender and receiver.
- Peer walks out of range.
- Screen off/lock during transfer.
- App backgrounded.
- Low battery and low storage.
- Wi-Fi already connected to access point.
- Mobile data on/off.
- Incoming phone interruption.
- Process death / app restart.
- Version mismatch.

## 5. Release signing

- Use Play App Signing for production distribution.
- Keep upload key in controlled encrypted storage with recovery procedure.
- Never embed signing key or credentials in repository/CI logs.
- Use separate debug, internal, beta and production build configurations as needed.
- Generate and archive release SBOM/dependency inventory, ProGuard mapping and exact source commit/tag.

## 6. Rollout

1. Internal testing track.
2. Closed beta with representative OEMs.
3. Fix crash/security/policy findings.
4. Staged production rollout beginning with a small percentage where available.
5. Monitor Play vitals/support reports and pause rollout on critical regressions.

## 7. Exit criteria

- No known critical/high security issue.
- 100% completed test transfers hash-verify.
- Required Play declarations and policy URLs live.
- Crash-free and ANR metrics acceptable in beta.
- No unresolved blocker on minimum supported OS/OEM matrix.
- Legal/operator placeholders resolved in public documents.